

Homework 11

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Problem 8.6

Exercise 8.6. Let X and Y be as in Exercise 8.1. Assume additionally that X and Y are independent. Evaluate all the expectations in part (d) of Exercise 8.1.

Exercise 8.1. Let n be a positive integer and p and r two real numbers in $(0, 1)$. Two random variables X and Y are defined on the same sample space. All we know about them is that $X \sim \text{Geom}(p)$ and $Y \sim \text{Bin}(n, r)$. For each expectation in parts (a)–(d) below, decide whether it can be calculated with this information, and if it can, give its value.

- (a) $E[X + Y]$
- (b) $E[XY]$
- (c) $E[X^2 + Y^2]$
- (d) $E[(X + Y)^2]$

Solution.

Scratch / Setup. We only need the expectation from part (d) of Exercise 8.1:

$$E[(X + Y)^2].$$

Expand:

$$E[(X + Y)^2] = E[X^2] + 2E[XY] + E[Y^2].$$

Because X and Y are independent,

$$E[XY] = E[X]E[Y].$$

For $X \sim \text{Geom}(p)$ on $\{1, 2, \dots\}$,

$$E[X] = \frac{1}{p}, \quad \text{Var}(X) = \frac{1-p}{p^2},$$

so

$$E[X^2] = \text{Var}(X) + (E[X])^2 = \frac{1-p}{p^2} + \frac{1}{p^2} = \frac{2-p}{p^2}.$$

For $Y \sim \text{Bin}(n, r)$,

$$E[Y] = nr, \quad \text{Var}(Y) = nr(1-r),$$

so

$$E[Y^2] = \text{Var}(Y) + (E[Y])^2 = nr(1-r) + n^2r^2.$$

We are computing

$$E[(X + Y)^2].$$

Using the identity

$$(X + Y)^2 = X^2 + 2XY + Y^2,$$

we get

$$E[(X + Y)^2] = E[X^2] + 2E[XY] + E[Y^2].$$

Since X and Y are independent,

$$E[XY] = E[X]E[Y] = \frac{1}{p} \cdot nr = \frac{nr}{p}.$$

Also,

$$E[X^2] = \frac{2-p}{p^2}, \quad E[Y^2] = nr(1-r) + n^2r^2.$$

Therefore

$$E[(X + Y)^2] = \frac{2-p}{p^2} + 2\frac{nr}{p} + nr(1-r) + n^2r^2.$$

Final: $E[(X + Y)^2] = \frac{2-p}{p^2} + \frac{2nr}{p} + nr(1-r) + n^2r^2$

Problem 8.12

Exercise 8.12.

(a) Let Z be $\text{Gamma}(2, \lambda)$ distributed. That is, Z has the density function

$$f_Z(z) = \begin{cases} \lambda^2 z e^{-\lambda z}, & z \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

Use the definition of the moment generating function to calculate $M_Z(t) = E[e^{tZ}]$.

(b) Let X and Y be two independent $\text{Exp}(\lambda)$ random variables. Recall the moment generating function of X and Y from Example 5.6 in Section 5.1. Using the approach from Section 8.3 show that Z and $X + Y$ have the same distribution.

Solution.

Scratch / Setup. For part (a), use the definition of the mgf:

$$M_Z(t) = E[e^{tZ}] = \int_0^\infty e^{tz} \lambda^2 z e^{-\lambda z} dz = \lambda^2 \int_0^\infty z e^{-(\lambda-t)z} dz.$$

This only makes sense for $t < \lambda$.

Now factor out $(\lambda - t)^{-2}$:

$$M_Z(t) = \frac{\lambda^2}{(\lambda - t)^2} \int_0^\infty (\lambda - t)^2 z e^{-(\lambda-t)z} dz.$$

The integrand is the $\text{Gamma}(2, \lambda - t)$ density, so the integral is 1.

For part (b), if $X, Y \sim \text{Exp}(\lambda)$ are independent, then

$$M_X(t) = M_Y(t) = \frac{\lambda}{\lambda - t}, \quad t < \lambda,$$

and hence

$$M_{X+Y}(t) = M_X(t)M_Y(t) = \left(\frac{\lambda}{\lambda - t} \right)^2.$$

(a) We compute the mgf directly:

$$M_Z(t) = \int_0^\infty e^{tz} \lambda^2 z e^{-\lambda z} dz = \lambda^2 \int_0^\infty z e^{-(\lambda-t)z} dz.$$

For $t < \lambda$, this becomes

$$M_Z(t) = \frac{\lambda^2}{(\lambda - t)^2} \int_0^\infty (\lambda - t)^2 z e^{-(\lambda-t)z} dz.$$

The integral is 1 because it is the total integral of a $\text{Gamma}(2, \lambda - t)$ density. Therefore

$$M_Z(t) = \left(\frac{\lambda}{\lambda - t} \right)^2, \quad t < \lambda.$$

(b) Since X and Y are independent $\text{Exp}(\lambda)$ random variables,

$$M_X(t) = M_Y(t) = \frac{\lambda}{\lambda - t}, \quad t < \lambda.$$

Hence

$$M_{X+Y}(t) = M_X(t)M_Y(t) = \left(\frac{\lambda}{\lambda - t}\right)^2.$$

Part (a) showed that this is exactly $M_Z(t)$ for $t < \lambda$. Since the moment generating functions agree on an interval containing 0, the distributions are the same. Thus

$$X + Y \sim \Gamma(2, \lambda).$$

Final: (a) $M_Z(t) = \left(\frac{\lambda}{\lambda - t}\right)^2$ for $t < \lambda$

Final: (b) $X + Y \sim \Gamma(2, \lambda)$

Problem 8.22

Exercise 8.22. In a lottery 5 numbers are chosen from the set $\{1, \dots, 90\}$ without replacement. We order the five numbers in increasing order and we denote by X the number of times the difference between two neighboring numbers is 1. (For example, for $\{1, 2, 3, 4, 5\}$ we have $X = 4$, for $\{3, 6, 8, 9, 24\}$ we have $X = 1$ and for $\{12, 14, 43, 75, 88\}$ we have $X = 0$.) Find $E[X]$.

Solution.

Scratch / Setup. It is cleaner to index the possible consecutive pairs of numbers rather than the four gaps in the ordered sample.

For $k = 1, \dots, 89$, let

$$J_k = \mathbf{1}\{\text{both } k \text{ and } k+1 \text{ are chosen}\}.$$

Then every selected consecutive pair contributes exactly 1 to X , so

$$X = \sum_{k=1}^{89} J_k.$$

For a fixed k ,

$$E[J_k] = P(J_k = 1) = \frac{\binom{88}{3}}{\binom{90}{5}},$$

since once k and $k+1$ are included, the remaining 3 chosen numbers can be any 3 of the other 88 numbers.

Let

$$J_k = \mathbf{1}\{\text{both } k \text{ and } k+1 \text{ are chosen}\}, \quad k = 1, \dots, 89.$$

Then

$$X = \sum_{k=1}^{89} J_k,$$

because X counts exactly how many consecutive integer pairs occur in the chosen 5-element set.

By linearity of expectation,

$$E[X] = \sum_{k=1}^{89} E[J_k] = \sum_{k=1}^{89} P(J_k = 1).$$

For a fixed k ,

$$P(J_k = 1) = \frac{\binom{88}{3}}{\binom{90}{5}}.$$

Therefore

$$E[X] = 89 \frac{\binom{88}{3}}{\binom{90}{5}}.$$

Now

$$\frac{\binom{88}{3}}{\binom{90}{5}} = \frac{88!}{3!85!} \cdot \frac{5!85!}{90!} = \frac{20}{90 \cdot 89} = \frac{2}{801}.$$

Hence

$$E[X] = 89 \cdot \frac{2}{801} = \frac{2}{9}.$$

Final: $E[X] = \frac{2}{9}$

Problem 8.23

Exercise 8.23. We have an urn with 20 red and 30 green balls. We draw all the balls one by one and record the colors we see. (That is, we take a sample of 50 without replacement.)

(a) Find the probability that the 28th and 29th balls are of different color.

(b) Let X be the number of times two consecutive balls are of different color. Find $E[X]$.

Solution.

Scratch / Setup. For part (a), by exchangeability any adjacent pair of positions has the same law as the first two draws.

For part (b), let

$$I_k = \mathbf{1}\{\text{balls } k \text{ and } k+1 \text{ have different colors}\}, \quad k = 1, \dots, 49.$$

Then

$$X = \sum_{k=1}^{49} I_k.$$

So

$$E[X] = \sum_{k=1}^{49} E[I_k] = 49 P(\text{two consecutive balls are different colors}).$$

(a) By exchangeability, the colors of the 28th and 29th draws have the same joint distribution as the colors of the first two draws. So

$$\begin{aligned} P(\text{28th and 29th different}) &= P(RG) + P(GR) \\ &= \frac{20}{50} \cdot \frac{30}{49} + \frac{30}{50} \cdot \frac{20}{49} \\ &= \frac{24}{49}. \end{aligned}$$

(b) Let

$$I_k = \mathbf{1}\{\text{balls } k \text{ and } k+1 \text{ have different colors}\}, \quad k = 1, \dots, 49.$$

Then

$$X = \sum_{k=1}^{49} I_k.$$

By linearity of expectation,

$$E[X] = \sum_{k=1}^{49} E[I_k] = \sum_{k=1}^{49} P(I_k = 1).$$

Again by exchangeability, $P(I_k = 1)$ is the same for every k , and part (a) shows that this common value is $24/49$. Therefore

$$E[X] = 49 \cdot \frac{24}{49} = 24.$$

Final: $\boxed{(a) \frac{24}{49}}$

Final: $\boxed{(b) E[X] = 24}$

Problem 8.36

Exercise 8.36. I roll a fair die four times. Let X be the number of different outcomes that I see. (For example, if the die rolls are 5, 3, 6, 6 then $X = 3$ because the different outcomes are 3, 5 and 6.)

(a) Find the mean of X .

(b) Find the variance of X .

Solution.

Scratch / Setup. For $j = 1, \dots, 6$, let

$$I_j = \mathbf{1}\{\text{face } j \text{ appears at least once in the four rolls}\}.$$

Then

$$X = I_1 + \dots + I_6.$$

For a fixed face j ,

$$P(I_j = 1) = 1 - P(\text{face } j \text{ never appears}) = 1 - \left(\frac{5}{6}\right)^4.$$

Call this probability p .

For $i \neq j$,

$$P(I_i = 1, I_j = 1) = 1 - P(i \text{ absent}) - P(j \text{ absent}) + P(i, j \text{ both absent}),$$

so

$$P(I_i = 1, I_j = 1) = 1 - 2\left(\frac{5}{6}\right)^4 + \left(\frac{4}{6}\right)^4.$$

Call this probability q .

Then

$$\text{Var}(X) = \sum_{j=1}^6 \text{Var}(I_j) + 2 \sum_{1 \leq i < j \leq 6} \text{Cov}(I_i, I_j).$$

(a) Let

$$I_j = \mathbf{1}\{\text{face } j \text{ appears at least once}\}, \quad j = 1, \dots, 6.$$

Then

$$X = I_1 + \dots + I_6.$$

By linearity of expectation,

$$E[X] = \sum_{j=1}^6 E[I_j] = \sum_{j=1}^6 P(I_j = 1).$$

For any fixed j ,

$$P(I_j = 1) = 1 - \left(\frac{5}{6}\right)^4.$$

Therefore

$$E[X] = 6 \left(1 - \left(\frac{5}{6}\right)^4\right) = \frac{671}{216}.$$

(b) Write

$$p = P(I_1 = 1) = 1 - \left(\frac{5}{6}\right)^4 = \frac{671}{1296}.$$

Then

$$\text{Var}(I_j) = p(1 - p).$$

For two distinct faces $i \neq j$,

$$\begin{aligned} q &= P(I_i = 1, I_j = 1) \\ &= 1 - 2 \left(\frac{5}{6}\right)^4 + \left(\frac{4}{6}\right)^4 = \frac{151}{648}, \end{aligned}$$

so

$$\text{Cov}(I_i, I_j) = q - p^2.$$

Since there are 6 diagonal terms and $\binom{6}{2} = 15$ distinct pairs,

$$\text{Var}(X) = 6p(1 - p) + 2\binom{6}{2}(q - p^2).$$

Therefore

$$\text{Var}(X) = 6p(1 - p) + 30(q - p^2) = \frac{20855}{46656}.$$

Final: (a) $E[X] = 6 \left(1 - \left(\frac{5}{6}\right)^4\right) = \frac{671}{216}$

Final: (b) $\text{Var}(X) = \frac{20855}{46656}$

Problem 8.40

Exercise 8.40. In a certain lottery 5 numbers are drawn without replacement out of the set $\{1, 2, \dots, 90\}$ randomly each week. Let X be the number of different numbers drawn during the first four weeks of the year. (For example, if the draws are $\{1, 2, 3, 4, 5\}$, $\{2, 3, 4, 5, 6\}$, $\{3, 4, 5, 6, 7\}$ and $\{4, 5, 6, 7, 8\}$ then $X = 8$.)

(a) Find $E[X]$.

Solution.

Scratch / Setup. For each number $m \in \{1, \dots, 90\}$, let

$$I_m = \mathbf{1}\{\text{number } m \text{ appears at least once in the first four weeks}\}.$$

Then

$$X = \sum_{m=1}^{90} I_m.$$

So

$$E[X] = \sum_{m=1}^{90} E[I_m] = 90 P(\text{a fixed number appears at least once}).$$

For a fixed number m , in one week

$$P(m \text{ is not drawn}) = \frac{\binom{89}{5}}{\binom{90}{5}} = \frac{17}{18}.$$

Since the weekly drawings are independent, the probability m is never drawn in the first four weeks is

$$\left(\frac{17}{18}\right)^4.$$

For each $m \in \{1, \dots, 90\}$, let

$$I_m = \mathbf{1}\{\text{number } m \text{ is drawn at least once during the first four weeks}\}.$$

Then

$$X = \sum_{m=1}^{90} I_m.$$

By linearity of expectation,

$$E[X] = \sum_{m=1}^{90} E[I_m] = 90 P(I_1 = 1).$$

Now fix a number, say 1. In one week,

$$P(1 \text{ is not drawn}) = \frac{\binom{89}{5}}{\binom{90}{5}} = \frac{17}{18}.$$

Because the four weekly drawings are independent,

$$P(1 \text{ is never drawn in the first four weeks}) = \left(\frac{17}{18}\right)^4.$$

Hence

$$P(I_1 = 1) = 1 - \left(\frac{17}{18}\right)^4.$$

Therefore

$$E[X] = 90 \left(1 - \left(\frac{17}{18}\right)^4\right).$$

Final: $E[X] = 90 \left(1 - \left(\frac{17}{18}\right)^4\right)$

Problem 8.42

Exercise 8.42. The random variables $X_1, \dots, X_n$ are i.i.d. We also know that $E[X_1] = 0$, $E[X_1^2] = a$, $E[X_1^3] = b$ and $E[X_1^4] = c$. Let $\bar{X}_n = \frac{X_1 + \dots + X_n}{n}$. Find the fourth moment of $\bar{X}_n$.

Solution.

Scratch / Setup. Let

$$S_n = X_1 + \dots + X_n.$$

Then

$$\bar{X}_n = \frac{S_n}{n}, \quad E[\bar{X}_n^4] = \frac{1}{n^4} E[S_n^4].$$

Expand S_n^4 by grouping terms according to exponent pattern:

$$S_n^4 = \sum_{i=1}^n X_i^4 + 4 \sum_{i \neq j} X_i^3 X_j + 6 \sum_{i < j} X_i^2 X_j^2 + 12 \sum_{\substack{i, j, k \\ \text{distinct}}} X_i^2 X_j X_k + 24 \sum_{i < j < k < \ell} X_i X_j X_k X_\ell.$$

Because the X_i are independent and $E[X_i] = 0$, any term containing a factor with odd first power has expectation 0. So only

$$\sum X_i^4 \quad \text{and} \quad \sum_{i < j} X_i^2 X_j^2$$

survive after taking expectation.

Let

$$S_n = X_1 + \dots + X_n.$$

Then

$$E[\bar{X}_n^4] = \frac{1}{n^4} E[S_n^4].$$

Expanding S_n^4 and taking expectations, all terms with an odd factor vanish because the X_i are independent and $E[X_i] = 0$. Thus only two types of terms remain:

$$E[S_n^4] = \sum_{i=1}^n E[X_i^4] + 6 \sum_{i < j} E[X_i^2 X_j^2].$$

By identical distribution,

$$E[X_i^4] = c$$

for every i , and by independence,

$$E[X_i^2 X_j^2] = E[X_i^2] E[X_j^2] = a^2 \quad (i \neq j).$$

Therefore

$$E[S_n^4] = nc + 6 \binom{n}{2} a^2 = nc + 3n(n-1)a^2.$$

Dividing by n^4 gives

$$E[\bar{X}_n^4] = \frac{nc + 3n(n-1)a^2}{n^4} = \frac{c + 3(n-1)a^2}{n^3}.$$

Final: $\boxed{E[\bar{X}_n^4] = \frac{c+3(n-1)a^2}{n^3}}$

Problem 8.48

Exercise 8.48. Suppose there are 100 cards numbered 1 through 100. Draw a card at random. Let X be the number of digits on the card (between 1 and 3) and let Y be the number of zeros. Find $\text{Cov}(X, Y)$.

Solution.

Scratch / Setup. Partition the numbers $1, \dots, 100$ into:

- 1-digit numbers: $1, \dots, 9$ (9 numbers),
- 2-digit numbers: $10, \dots, 99$ (90 numbers),
- the 3-digit number: 100.

For Y :

- among $10, \dots, 99$, only $10, 20, \dots, 90$ contain a zero, each with one zero,
- 100 has two zeros.

So we can compute $E[X]$, $E[Y]$, and $E[XY]$ by averaging over these cases, then use

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y].$$

We compute

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y].$$

First,

$$E[X] = \frac{1}{100}(9 \cdot 1 + 90 \cdot 2 + 1 \cdot 3) = \frac{192}{100} = \frac{48}{25}.$$

Next, the total number of zeros among the integers $1, \dots, 100$ is

$$9 \cdot 1 + 2 = 11,$$

since $10, 20, \dots, 90$ contribute one zero each and 100 contributes two zeros. Hence

$$E[Y] = \frac{11}{100}.$$

For $E[XY]$, the only numbers with $Y > 0$ are $10, 20, \dots, 90$ and 100. Thus

$$E[XY] = \frac{1}{100}(9 \cdot (2 \cdot 1) + 1 \cdot (3 \cdot 2)) = \frac{18 + 6}{100} = \frac{24}{100} = \frac{6}{25}.$$

Therefore

$$\text{Cov}(X, Y) = \frac{6}{25} - \frac{48}{25} \cdot \frac{11}{100} = \frac{600 - 528}{2500} = \frac{72}{2500} = \frac{18}{625}.$$

Final: $\boxed{\text{Cov}(X, Y) = \frac{18}{625}}$

Problem 8.54

Exercise 8.54. Suppose that for the random variables X, Y we have $E[X] = 2$, $E[Y] = 1$, $E[X^2] = 5$, $E[Y^2] = 10$ and $E[XY] = 1$.

(a) Compute $\text{Corr}(X, Y)$.

(b) Find a number c so that X and $X + cY$ are uncorrelated.

Solution.

Scratch / Setup. We are given

$$E[X] = 2, \quad E[Y] = 1, \quad E[X^2] = 5, \quad E[Y^2] = 10, \quad E[XY] = 1.$$

So

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 1 - 2 \cdot 1 = -1.$$

Also,

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 5 - 4 = 1,$$

$$\text{Var}(Y) = E[Y^2] - (E[Y])^2 = 10 - 1 = 9.$$

For part (b),

$$\text{Cov}(X, X + cY) = \text{Cov}(X, X) + c \text{Cov}(X, Y) = \text{Var}(X) + c \text{Cov}(X, Y).$$

Set this equal to 0 and solve for c .

(a) We first compute the covariance:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 1 - 2 \cdot 1 = -1.$$

Next,

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 5 - 4 = 1,$$

and

$$\text{Var}(Y) = E[Y^2] - (E[Y])^2 = 10 - 1 = 9.$$

Therefore

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}} = \frac{-1}{\sqrt{1 \cdot 9}} = -\frac{1}{3}.$$

(b) We want X and $X + cY$ to be uncorrelated, so we set

$$\text{Cov}(X, X + cY) = 0.$$

By linearity of covariance,

$$\text{Cov}(X, X + cY) = \text{Cov}(X, X) + c \text{Cov}(X, Y) = \text{Var}(X) + c \text{Cov}(X, Y).$$

Using the values from part (a),

$$0 = 1 + c(-1) = 1 - c,$$

so

$$c = 1.$$

Final: (a) $\text{Corr}(X, Y) = -\frac{1}{3}$

Final: (b) $c = 1$